HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed production-grade C++ applications for real-time data processing, contributing to a **15% reduction** in end-to-end pipeline latency.
- Implemented Python scripts to automate data ingestion and pre-processing for machine learning model training, handling datasets up to **500GB**.
- Engineered **Linux shell scripts** for system monitoring and log analysis, improving incident detection time by approximately **20%**.
- Built and optimized **C++ algorithms** for image classification tasks, achieving a **92% accuracy** on a benchmark dataset.
- Developed Python-based machine learning solutions for defect detection using **PyTorch**, integrating with embedded hardware for edge deployment.
- Contributed to the design and implementation of **Python and C++** production software, participating actively in code reviews and technical documentation.
- Containerised C++ and Python microservices using **Docker**, deploying to a cloud environment for scalable inference and testing.

SKILLS

Languages – C++, Python, Linux Shell Scripting

Machine Learning – PyTorch, TensorFlow, Algorithms, Deep Learning, Defect Detection, Classification, Measurement

Development Tools – Git, Jenkins, Docker, CMake

Databases – SQL

Operating Systems – Linux

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

PROJECTS

C++ Computer Vision System for Defect Analysis

- Developed a real-time C++ application to analyze images from embedded hardware, identifying and classifying defects with **90% precision**.
- Integrated **OpenCV** and custom algorithms within the C++ codebase to perform image pre-processing and feature extraction.
- Optimized algorithm performance on edge devices, reducing processing time per image by **30%** through C++ code refactoring.

Python ML Pipeline for Predictive Maintenance

- Engineered a Python data pipeline to extract features from sensor logs for predictive maintenance models, processing approximately **3TB of historical data**.
- Trained a **scikit-learn** regression model to predict component failure probability, achieving an RMSE of **0.12** on a held-out test set.
- Developed REST APIs using **FastAPI** to serve model predictions, integrating with an existing enterprise monitoring platform.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)